

$$1) |a\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad |b\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad |c\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$|\psi\rangle = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{i\sqrt{7}}{\sqrt{15}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \quad \langle\psi| = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{i\sqrt{7}}{\sqrt{15}} & \frac{1}{\sqrt{3}} \end{bmatrix} \quad \langle\psi|\psi\rangle = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{i\sqrt{7}}{\sqrt{15}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{i\sqrt{7}}{\sqrt{15}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} = [1]$$

$|\psi\rangle$ is state normalized because $\langle\psi|\psi\rangle = 1$

$$2) X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0| \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = i|1\rangle\langle 0| - i|0\rangle\langle 1|$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1| \quad \langle 1|0\rangle = \langle 0|1\rangle = 0 \quad \langle 0|0\rangle = \langle 1|1\rangle = 1$$

$$\begin{aligned} X^2 &= (|0\rangle\langle 1| + |1\rangle\langle 0|) \cdot (|0\rangle\langle 1| + |1\rangle\langle 0|) \\ &= |0\rangle\langle 1| \underbrace{|0\rangle\langle 1|}_0 + |0\rangle\langle 1| \underbrace{|1\rangle\langle 0|}_1 + |1\rangle\langle 0| \underbrace{|0\rangle\langle 1|}_1 + |1\rangle\langle 0| \underbrace{|1\rangle\langle 0|}_0 \\ &= |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbb{1} \end{aligned}$$

$$\begin{aligned} Y^2 &= (i|1\rangle\langle 0| - i|0\rangle\langle 1|) \cdot (i|1\rangle\langle 0| - i|0\rangle\langle 1|) \\ &= -\underbrace{|1\rangle\langle 0| |1\rangle\langle 0|}_0 + \underbrace{|1\rangle\langle 0| |0\rangle\langle 1|}_1 + \underbrace{|0\rangle\langle 1| |1\rangle\langle 0|}_1 - \underbrace{|0\rangle\langle 1| |0\rangle\langle 1|}_0 \\ &= |1\rangle\langle 1| + |0\rangle\langle 0| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbb{1} \end{aligned}$$

$$\begin{aligned} Z^2 &= (|0\rangle\langle 0| - |1\rangle\langle 1|) \cdot (|0\rangle\langle 0| - |1\rangle\langle 1|) \\ &= \underbrace{|0\rangle\langle 0| |0\rangle\langle 0|}_1 - \underbrace{|0\rangle\langle 0| |1\rangle\langle 1|}_0 - \underbrace{|1\rangle\langle 1| |0\rangle\langle 0|}_0 + \underbrace{|1\rangle\langle 1| |1\rangle\langle 1|}_1 \\ &= |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbb{1} \end{aligned}$$

$$3) A = i|1\rangle\langle 1| + \frac{\sqrt{3}}{2}|1\rangle\langle 2| + 2|2\rangle\langle 1| - |2\rangle\langle 3|$$

$$|1\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad |2\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad |3\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

040220506
Emir Ali Kara

$$|1\rangle\langle 1| = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|1\rangle\langle 2| = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|2\rangle\langle 1| = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|2\rangle\langle 3| = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = i \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \frac{\sqrt{3}}{2} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} i & \frac{\sqrt{3}}{2} & 0 \\ 2 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$a) A^\dagger = \begin{bmatrix} -i & 2 & 0 \\ \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

b) If $A^\dagger = A^{-1}$ then A is unitary
 A is not unitary because A is not a square matrix which A is do not have inverse matrix

$$4) [A, B]^\dagger \stackrel{?}{=} [B^\dagger, A^\dagger]$$

$$[A, B]^\dagger = (AB - BA)^\dagger = (B^\dagger A^\dagger - A^\dagger B^\dagger) = [B^\dagger, A^\dagger]$$

$$5) [\sigma_x \sigma_y, \sigma_z] \stackrel{?}{=} 0$$

$$[\sigma_x \sigma_y, \sigma_z] = \sigma_x \sigma_y \sigma_z - \sigma_z \sigma_x \sigma_y$$

$$\sigma_x \sigma_y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = i \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = i \sigma_z$$

$$[\sigma_x \sigma_y, \sigma_z] = i \sigma_z \sigma_z - \sigma_z \cdot i \sigma_z = 0$$

$$6) A|\psi\rangle = \alpha|\psi\rangle, \quad B|\psi\rangle = |\phi\rangle$$

$$A(B|\psi\rangle) = A|\phi\rangle \xrightarrow{\text{since } A \text{ and } B \text{ commute}} B(A|\psi\rangle) = B(\alpha|\psi\rangle) = \alpha B|\psi\rangle = \alpha|\phi\rangle$$

$$B(\alpha|\psi\rangle) = \alpha B|\psi\rangle \Rightarrow \alpha(B|\psi\rangle) = \alpha|\phi\rangle$$

$$7) A = \begin{bmatrix} 33 & 24 \\ 48 & 57 \end{bmatrix}$$

040220506
Emir Ali Kasa
Emir

$$\det \begin{pmatrix} 33-\lambda & 24 \\ 48 & 57-\lambda \end{pmatrix} = 0 \Rightarrow \det \left(3 \begin{pmatrix} 11-\lambda/3 & 8 \\ 16 & 19-\lambda/3 \end{pmatrix} \right) = 0$$

$$9 \det \begin{pmatrix} 11-\lambda/3 & 8 \\ 16 & 19-\lambda/3 \end{pmatrix} = 0 \quad \frac{\lambda^2}{9} - 10\lambda + 81 = 0 \Rightarrow \begin{matrix} \lambda^2 & -30\lambda & +81 \cdot 9 \\ 2 & -9 & -81 \\ 2 & & \end{matrix}$$

$$\lambda_1 = 9 \Rightarrow \begin{bmatrix} 24 & 24 \\ 48 & 48 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0 \quad |\psi_1\rangle = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda_2 = 81 \Rightarrow \begin{bmatrix} -48 & 24 \\ 48 & -24 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 0 \quad |\psi_2\rangle = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \quad D = \begin{bmatrix} 9 & 0 \\ 0 & 81 \end{bmatrix} \quad S^{-1} = \frac{1}{\det S} \text{Adj} S$$

$$\det S = 3 \quad \text{Adj} S = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

$$S^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{bmatrix}$$

$$A = SDS^{-1} \Rightarrow A^n = SD^nS^{-1}$$

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 9 & 0 \\ 0 & 81 \end{bmatrix} \begin{bmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{bmatrix}$$

$$A^{1/2} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{bmatrix}$$

$$A^{1/2} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 3 \end{bmatrix}$$

$$A^{1/2} = \begin{bmatrix} 5 & 2 \\ 4 & 7 \end{bmatrix}$$